Huan Wang

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Biography

I am a third-year PhD candidate at the University of Wollongong (UOW) under the supervision of Prof. Jun Shen and Prof. Jun Yan. I also work closely with Prof. Guansong Pang at Singapore Management University (SMU). Previously, I received my Master's degree at Xidian University (China) under the supervision of Prof. Lijuan Wang, and my Bachelor's degree from Anhui University of Science and Technology (China).

Recent Publications

- **Huan Wang**, et al., Guansong Pang. "*Domain-Skewed Federated Learning with Feature Decoupling and Calibration*". In: **CVPR'26** (CORE A*). June 3 - June 7, 2026, Denver.
- **Huan Wang**, et al., Jun Shen. "*Towards Zero-Shot Diabetic Retinopathy Grading: Learning Generalized Knowledge via Prompt-Driven Matching and Emulating*". In: **AAAI'26 oral** (CORE A*), 2026, Singapore.
- **Huan Wang**, et al., Jun Shen. "*FedDifRC: Unlocking the Potential of Text-to-Image Diffusion Models in Heterogeneous Federated Learning*". In: **ICCV'25** (CORE A*). Oct 19 - Oct 23, 2025, Honolulu.
- **Huan Wang**, et al., Jun Shen. "*FedSC: Federated Learning with Semantic-Aware Collaboration*". In: **ACM KDD'25 oral** (CORE A*). August 3 - August 7, 2025, Toronto.
- **Huan Wang**, et al., Jun Shen. "*Dual Visual Prompting with Context-Modulated Diffusion Prompts*". In: **IEEE Transactions on Multimedia** (CORE A*), 2025.
- Haoran Li*, Xingjian Li*, **Huan Wang***, et al., Min Xu. "*3D Domain Adaptation for Cryo-Electron Tomography Segmentation*". In: **International Journal of Computer Vision** (CORE A*), 2026.

Research Interests

My research areas focus on leveraging deep learning techniques to build **Trustworthy AI**:

1. **Open-Set Visual Generalized Anomaly Detection**: To address practical challenges in open-world environments, including unknown categories, distribution shifts, and diverse anomaly patterns, I study how to enable models to generalize robustly to unseen, cross-domain, and semantically complex anomalies.
2. **Heterogeneous Federated Learning**: I focus on developing efficient and robust heterogeneous FL methods to address non-IID data and client-side model heterogeneity. In particular, my works mainly explore knowledge decoupling and transfer to improve model generalizability, fairness, and stability while preserving privacy.
3. **Prompt Tuning with Large Foundation Models**: I focus on developing efficient prompt tuning methods to enhance the adaptation of the large foundation models to downstream tasks, particularly for achieving stronger understanding and perception in few-shot, zero-shot, and out-of-distribution scenarios.

Services and Awards

- **Program Committee (Selected)**: ICCV, CVPR, NeurIPS, KDD, AAAI, ICML, ECCV, ICLR
- **Journal Reviewer (Selected)**: IEEE Transactions on Image Processing (TIP), IEEE Transactions on Multimedia (TMM), IEEE Transactions on Neural Networks and Learning Systems (TNNLS), IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- **Student Travel Award**, International Conference on Computer Vision (ICCV), 2025, Honolulu
- **Student Travel Award**, ACM SIGKDD (KDD), 2025, Toronto
- **Outstanding Program Committee**, AAAI Conference on Artificial Intelligence (AAAI), 2026, Singapore
- **Outstanding Reviewer**, Computer Vision and Pattern Recognition Conference (CVPR), 2025, Nashville

Education

- **Singapore Management University**: *Information Science (**Visiting PhD**)* 12/2025 – 01/2027
- **University of Wollongong**: *Information Science (**PhD**)* 02/2024 – 06/2027
- **Xidian University**: *Computer Science (**Master**)* 09/2020 – 06/2023
- **Anhui University of Science and Technology**: *Computer Science (**Bachelor**)* 09/2016 – 06/2020